

LOUIE CAI

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EXPERIENCE

Software Engineer

Feb 2025 – Present

Stripe

- Build software infrastructure, including creation, modification, and testing of source code in Ruby, JavaScript, and JVM languages.
- Monitor production web services for abnormal behavior and respond to incidents.

Software Development Engineer Intern

Jun – Sept 2024

Amazon

- Developed a full-stack portal with React front-end and AWS serverless back-end (Lambda, Step Functions, DynamoDB), automating internal experiment creation and tracking — cut deployment time from 2–3 days to minutes.
- Built unified portal enabling experiment creation and access to historical data.
- Managed infrastructure with AWS CDK and CloudFormation for seamless deployment and monitoring.

Software Development Engineer Intern

Jun – Sept 2023

Amazon

- Built a React + TypeScript internal tool that streamlined backend database onboarding, reducing the process from 2–3 days to several hours.
- Collaborated in an Agile team to deliver responsive, user-friendly interfaces.

CSE Tutor

Sept 2023 – Dec 2024

UC San Diego

- Held office hours for CSE 8A and CSE 140L; assisted students with coursework.
- Automated calculation of class grades with Python scripts.
- Performed TA duties: designed and created programming assignments and exams.

EDUCATION

University of California, San Diego, Data Science B.S.

Graduation: Dec 2024

- **GPA:** 3.7
- **Minor:** Computer Engineering
- **Relevant Coursework:** Machine Learning, Deep Learning, NLP, Data Analysis, Computer Organization and Systems Programming, Data Structures & Object-Oriented Design, Differential Equations, Linear Algebra, Probability and Statistics, Discrete Mathematics, Digital Systems Design

PROJECTS

Minecraft Netherite Horse Armor Mod ([CurseForge](#)) ([GitHub](#))

Jun 2020

- Organically grew to over **2.8 million downloads** on CurseForge.
- Built with Java and Minecraft Forge; extended vanilla gameplay by adding Netherite Horse Armor.

Twitter Sentiment Analysis API ([GitHub](#))

Jul 2022

- Developed a configurable LSTM model for tweet sentiment classification, served via FastAPI.
- Trained 1,000 LSTM variants with grid search to find optimal hyperparameters.

ACTIVITIES

IEEE F1TENTH IV2023 Robotics Championship - Anchorage ([Website](#))

Jun 2023

- Constructed two autonomous 1/10-scale RC cars equipped with LiDAR sensors and cameras for precise racetrack navigation.
- Led development of advanced navigation algorithms using ROS2 and DonkeyCar frameworks.
- Represented UC San Diego at the international championship — placed **4th**.

SKILLS

- **Languages:** Python, TypeScript, JavaScript, Ruby, Java, C, C++, C#, R, SQL, Bash, SystemVerilog

- **Web:** React, Vite, FastAPI, Spring Boot, HTML, CSS, React Hook Form, React Router DOM
- **ML / Data:** PyTorch, pandas, NumPy, scikit-learn, matplotlib, Selenium, Jupyter
- **Cloud / DevOps:** AWS Lambda, AWS Step Functions, AWS DynamoDB, AWS CDK, AWS CloudFormation, Docker
- **Tools:** Git, Linux, Vim, Unity, Blender, Gradle, JavaFX, Valgrind, Verilator